

## REDWOOD CITY, CA, USA

WMO: 994041

Lat: 37.5068N

Lon: 122.2119W

Elev: 20

StdP: 14.69

Time Zone: -8.00 (NAP)

Period: 05-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 12            | 38.6       | 40.7 | N/A                           | N/A | N/A  | N/A | N/A | N/A  | 23.1                  | 53.4 | 19.8 | 53.4 | 2.4                      | 150  | 0.470 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |     |      |     |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|-----|------|-----|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%  |      | 2%  |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB  | MCDB | WB  | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k) | (l)  | (m) | (n)  | (o)                  | (p)  |
| 9             | 14.1                   | 84.0            | N/A  | 79.7 | N/A  | 76.2 | N/A  | N/A                 | N/A  | N/A | N/A  | N/A | N/A  | 9.4                  | 320  |

## Dehumidification DP/MCDB and HR

| 0.4% |     |      | 1%  |     |      | 2%  |     |      | 0.4% |      | 1%   |      | 2%   |      | Extreme<br>Max WB |
|------|-----|------|-----|-----|------|-----|-----|------|------|------|------|------|------|------|-------------------|
| DP   | HR  | MCDB | DP  | HR  | MCDB | DP  | HR  | MCDB | Enth | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)  | (b) | (c)  | (d) | (e) | (f)  | (g) | (h) | (i)  | (j)  | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| N/A  | N/A | N/A  | N/A | N/A | N/A  | N/A | N/A | N/A  | N/A  | N/A  | N/A  | N/A  | N/A  | N/A  | N/A               |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |       |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|-------|----|----|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |       | DB | WB |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max   |    |    |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)   |    |    |
| 22.6              | 19.2 | 17.2 | 34.5                       | 92.3 | 2.3                | 3.1 | 32.8                                               | 94.5 | 31.5       | 96.4 | 30.2       | 98.1 | 28.5       | 100.4 |    |    |
|                   |      |      | N/A                        | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A   |    |    |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 58.6   | 51.3  | 52.7  | 55.0  | 57.0  | 59.6  | 63.3  | 64.8  | 65.1  | 65.3  | 62.3  | 55.9  | 50.9 |
|                                                                               | DBStd       | 6.88   | 4.06  | 4.59  | 4.54  | 4.97  | 4.57  | 4.64  | 3.64  | 3.64  | 4.52  | 4.50  | 4.53  | 4.65 |
|                                                                               | HDD50       | 118    | 34    | 23    | 7     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 8     | 45   |
|                                                                               | HDD65       | 2556   | 425   | 345   | 310   | 247   | 184   | 86    | 46    | 43    | 47    | 111   | 275   | 437  |
|                                                                               | CDD50       | 3266   | 74    | 98    | 163   | 212   | 296   | 399   | 460   | 467   | 460   | 380   | 184   | 73   |
|                                                                               | CDD65       | 231    | 0     | 0     | 1     | 8     | 15    | 36    | 41    | 45    | 58    | 26    | 1     | 0    |
|                                                                               | CDH74       | 1241   | 0     | 0     | 13    | 82    | 136   | 234   | 151   | 165   | 299   | 155   | 6     | 0    |
|                                                                               | CDH80       | 329    | 0     | 0     | 0     | 13    | 37    | 71    | 40    | 48    | 91    | 29    | 0     | 0    |
| Wind                                                                          | WSAvg       | 6.5    | 3.9   | 5.7   | 6.4   | 8.3   | 9.5   | 8.9   | 8.2   | 7.5   | 6.5   | 5.0   | 4.2   | 4.3  |
| Precipitation                                                                 | PrecAvg     | 19.6   | 3.9   | 3.6   | 2.7   | 1.3   | 0.7   | 0.2   | 0.0   | 0.0   | 0.1   | 1.0   | 2.2   | 4.0  |
|                                                                               | PrecMax     | 33.8   | 9.4   | 12.7  | 7.6   | 4.1   | 2.6   | 1.6   | 0.1   | 0.8   | 0.4   | 3.7   | 6.6   | 9.7  |
|                                                                               | PrecMin     | 7.8    | 0.3   | 0.0   | 0.1   | 0.1   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.2  |
|                                                                               | PrecStd     | 5.6    | 2.6   | 2.5   | 2.2   | 1.0   | 0.7   | 0.4   | 0.0   | 0.2   | 0.1   | 1.0   | 1.6   | 2.7  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 68.0  | 71.5  | 75.8  | 82.3  | 84.7  | 88.8  | 85.7  | 87.5  | 88.5  | 84.0  | 74.5  | 65.2 |
|                                                                               |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               | 2%          | DB     | 63.7  | 67.0  | 69.5  | 75.9  | 77.5  | 81.0  | 77.7  | 78.0  | 82.2  | 78.1  | 69.7  | 62.5 |
|                                                                               |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               | 5%          | DB     | 60.6  | 63.3  | 65.6  | 69.3  | 70.9  | 75.2  | 73.5  | 74.1  | 76.7  | 74.2  | 66.2  | 60.3 |
|                                                                               |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               | 10%         | DB     | 58.2  | 60.3  | 62.5  | 64.6  | 66.4  | 70.7  | 70.9  | 71.1  | 72.7  | 70.2  | 63.2  | 58.5 |
|                                                                               |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               | 2%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               | 5%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               | 10%         | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 12.7  | 12.5  | 13.2  | 13.3  | 12.7  | 14.0  | 12.7  | 12.8  | 14.1  | 14.7  | 13.4  | 11.6 |
|                                                                               |             | MCDBR  | 17.1  | 18.3  | 18.5  | 21.8  | 21.5  | 22.5  | 17.1  | 17.7  | 22.3  | 22.1  | 18.3  | 12.9 |
|                                                                               |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               | 5% WB       | MCDDBR | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.324  | 0.333 | 0.339 | 0.357 | 0.367 | 0.360 | 0.366 | 0.345 | 0.350 | 0.337 | 0.321 | 0.324 |      |
|                                                                               | taud        | 2.553  | 2.525 | 2.499 | 2.417 | 2.412 | 2.438 | 2.430 | 2.484 | 2.456 | 2.535 | 2.561 | 2.558 |      |
|                                                                               | Ebn at Noon | 275    | 286   | 294   | 293   | 290   | 291   | 288   | 293   | 286   | 280   | 273   | 266   |      |
|                                                                               | Edh at Noon | 24     | 28    | 31    | 36    | 37    | 36    | 36    | 33    | 32    | 27    | 24    | 23    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 768    | 1064  | 1504  | 1959  | 2235  | 2434  | 2277  | 2032  | 1735  | 1293  | 902   | 674   |      |
|                                                                               | RadStd      | 101    | 125   | 129   | 152   | 147   | 102   | 78    | 94    | 68    | 80    | 56    | 94    |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Regional (44 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
|                         | N/A   | N/A     | -2.11  | N/A     | N/A   | N/A   | N/A         | -198  | +243  | N/A   |

Nomenclature: See separate page